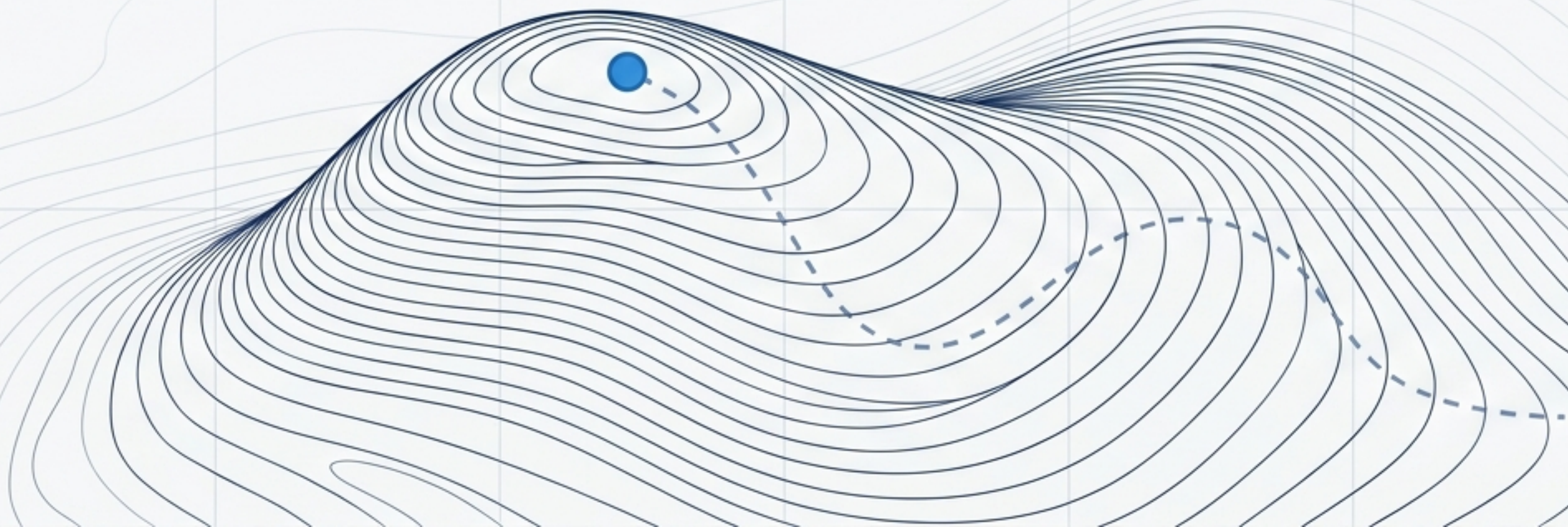


Topography of Meaning

Document Perturbation as Constrained Navigation on a Semantic Manifold

Coritense as a Semantic Marifold

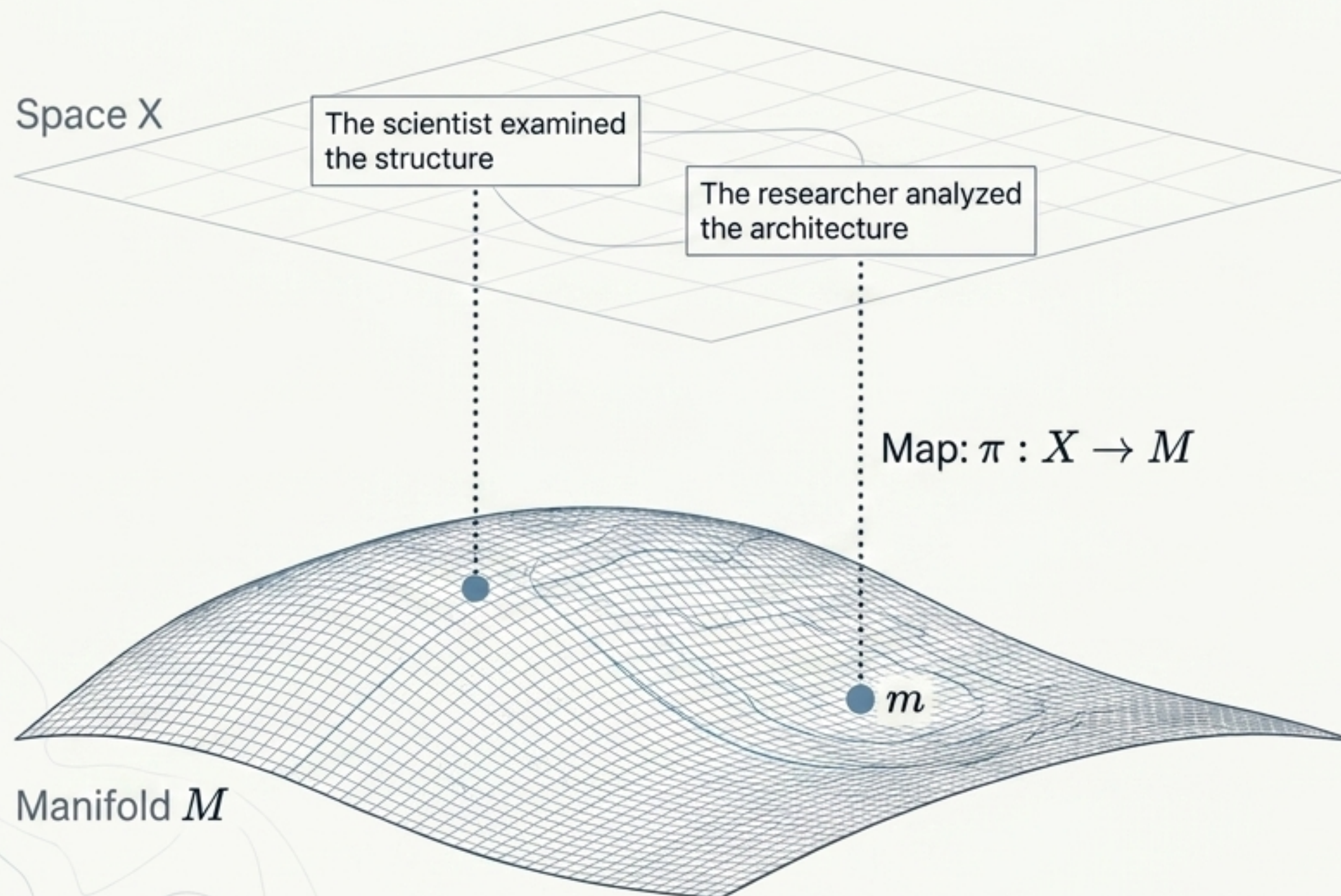
A geometric framework for formalizing semantic drift, conservation laws, and semantic colored noise.



Three eras of document augmentation

	Lexical Substitution (WordNet)	Unconstrained LLM (GPT)	Geometric Constrained (DocPerturb)
Ontological View	Document as String	Document as Distribution	Document as Geometric Coordinate
Contextual Fit	Fails (Produces Synonym Salad) ✗	Excellent (Fluency optimized)	Guaranteed by Manifold Topology ✓
Semantic Drift Control	None (Equiprobable sampling)	None (Invisible semantic drift)	Formal bounds via Distance Budget (B)
Non-Holonomic Protection	Fails on negation/causality	Hallucinates logical inversions	Protected via Semantic Connection

The text is secondary; the geometric object is primary



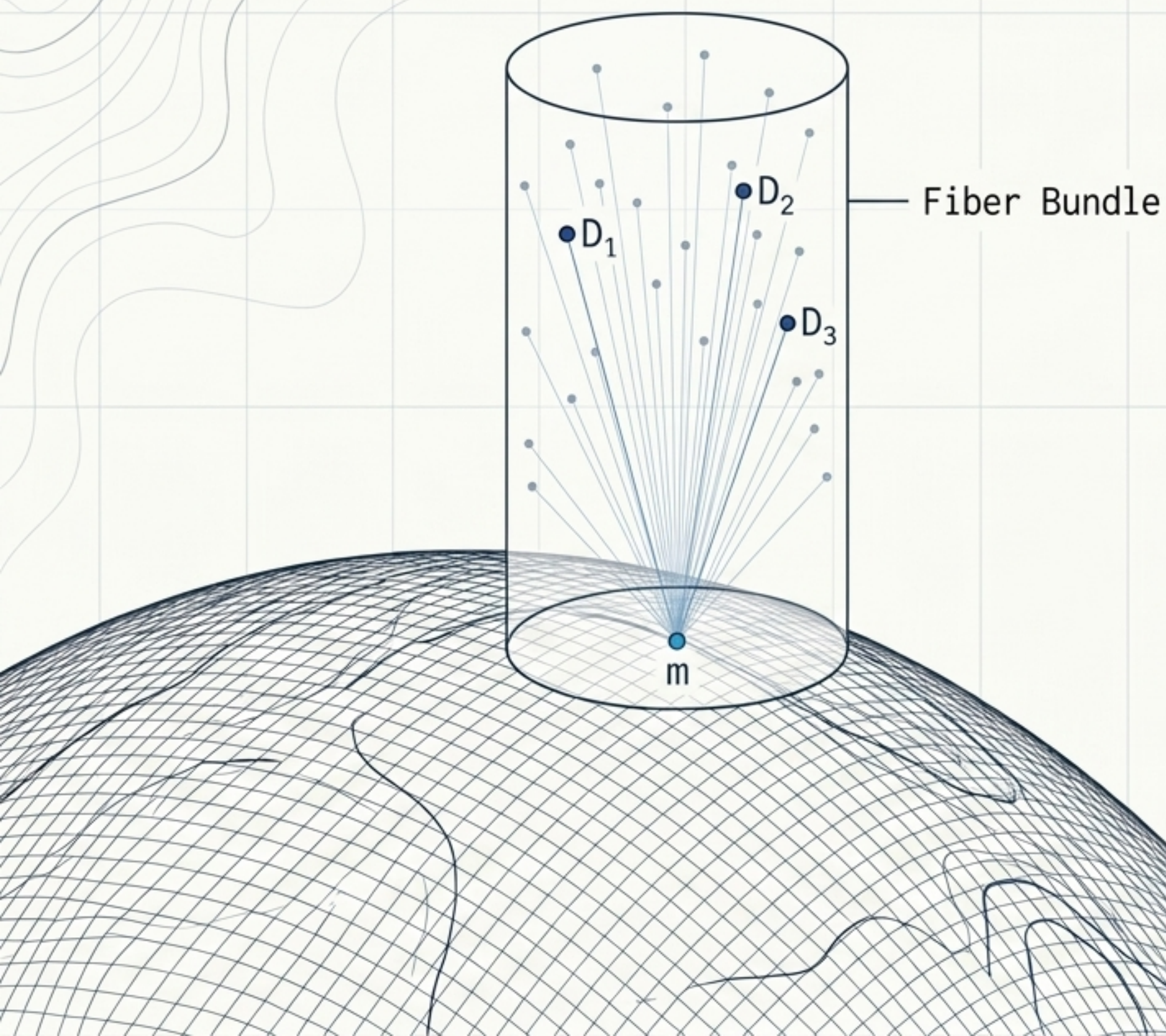
Space X :
The space of all syntactically valid documents (strings).

Projection π :
The map from surface realization to semantic identity.

Manifold M :
The underlying geometric reality where true meaning resides.

Core Insight: We are no longer asking which words to replace.
We are asking which directions in semantic space are accessible.

Document augmentation is exploration of the semantic fiber



Semantic Fiber $\pi^{-1}(m)$:

The exact set of documents that project down to identical meaning.

Approximate Fiber:

Paraphrases operating at a specific tolerance ϵ , forming an equivalence class.

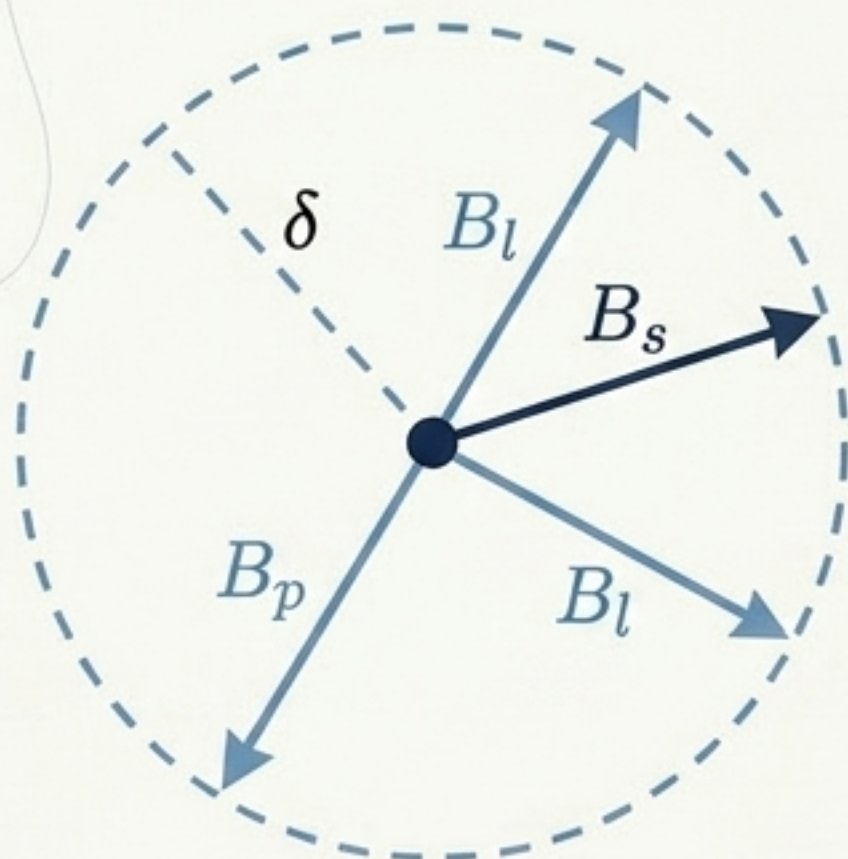
Reachable Set $R_B(D)$:

The precise volume of semantic space that is accessible under a given budget B .

Takeaway: The original document D_0 is not privileged. It is merely a choice of section—one canonical realization plucked from the fiber.

Semantic budgets constrain distance; curvature constraints prevent reversals

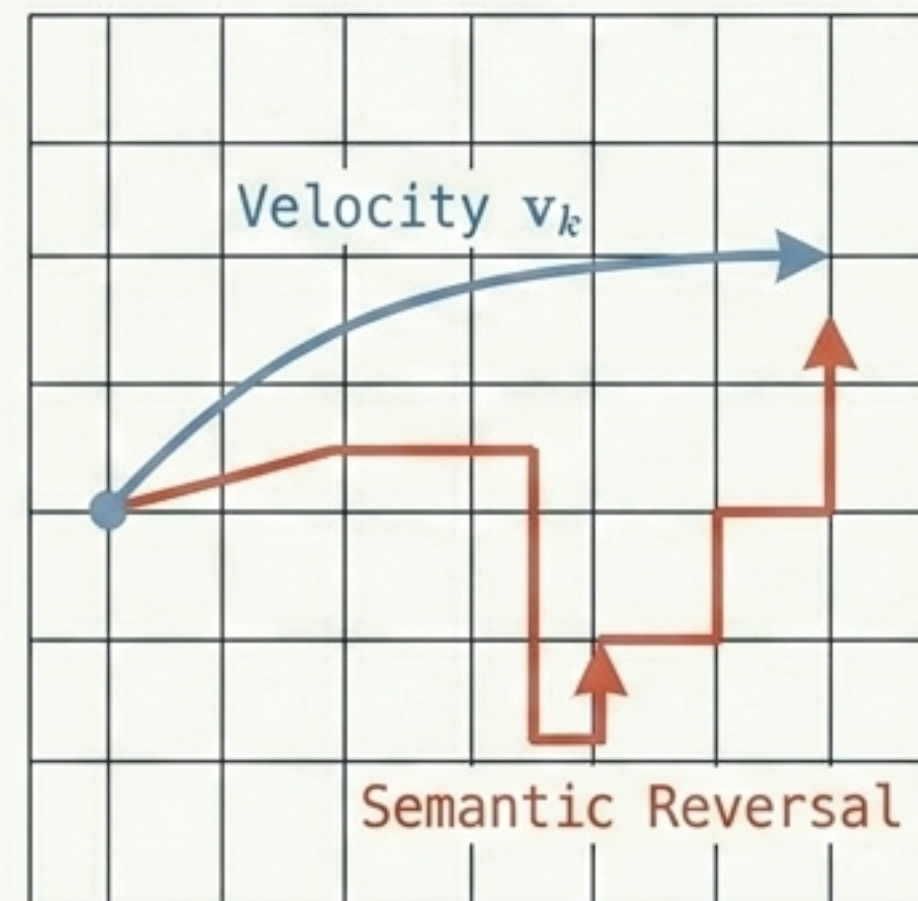
The Budget Vector ($\mathbf{B}$)



$$\mathbf{B} = (B_l, B_s, B_p, \delta)$$

Allocates capacity for lexical, syntactic, and semantic motion, plus maximum permitted step size.

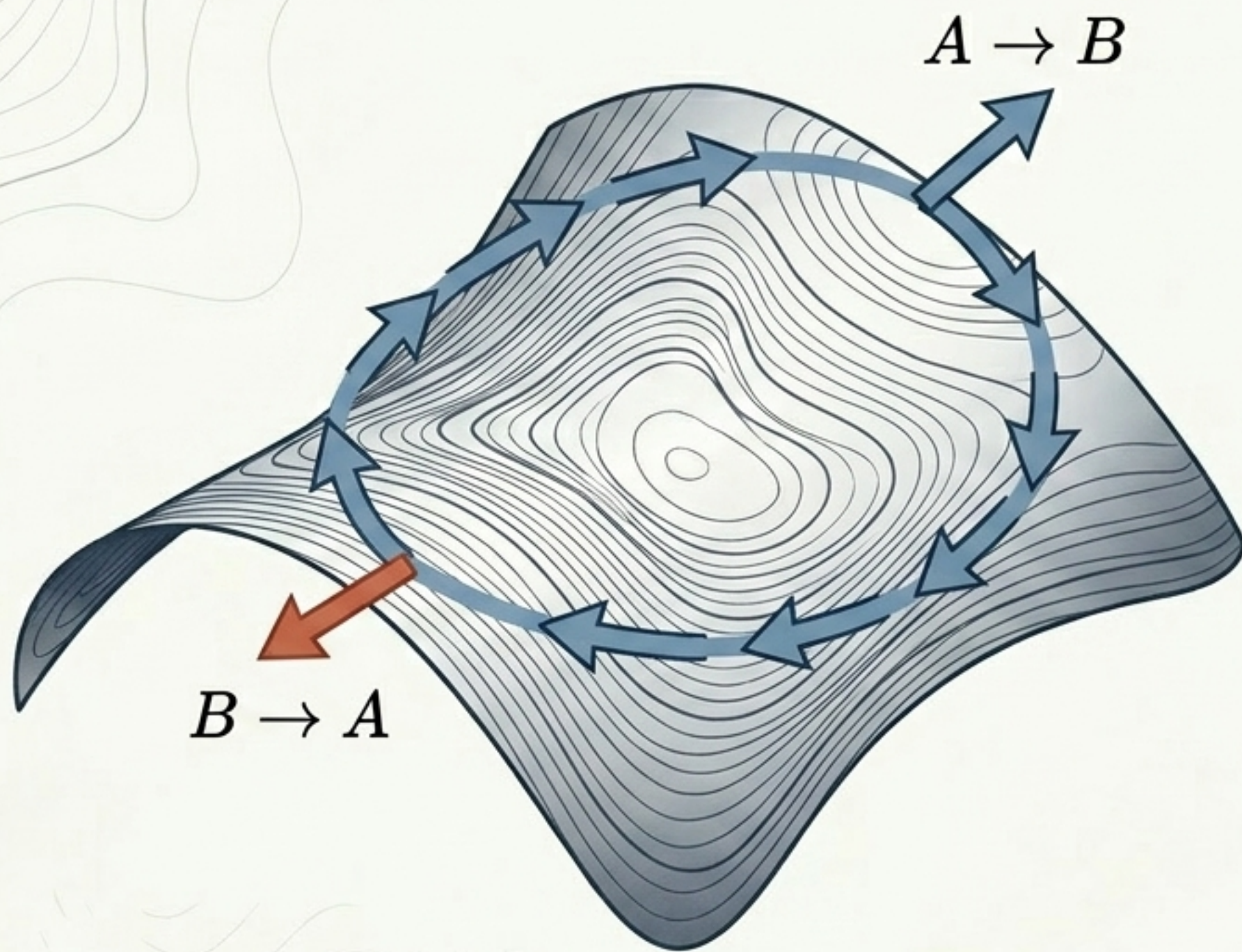
Trajectory Curvature (κ_k)



$$\kappa_k = 1 - \frac{\langle v_k, v_{k+1} \rangle}{\|v_k\| \|v_{k+1}\|}$$

Unconstrained bounds fail: Consecutive orthogonal steps accumulate distance but can return to the origin. Bounding $\kappa_k \leq \kappa_{max}$ physically prevents sharp semantic reversals.

The Semantic Connection (∇) prevents invisible semantic drift



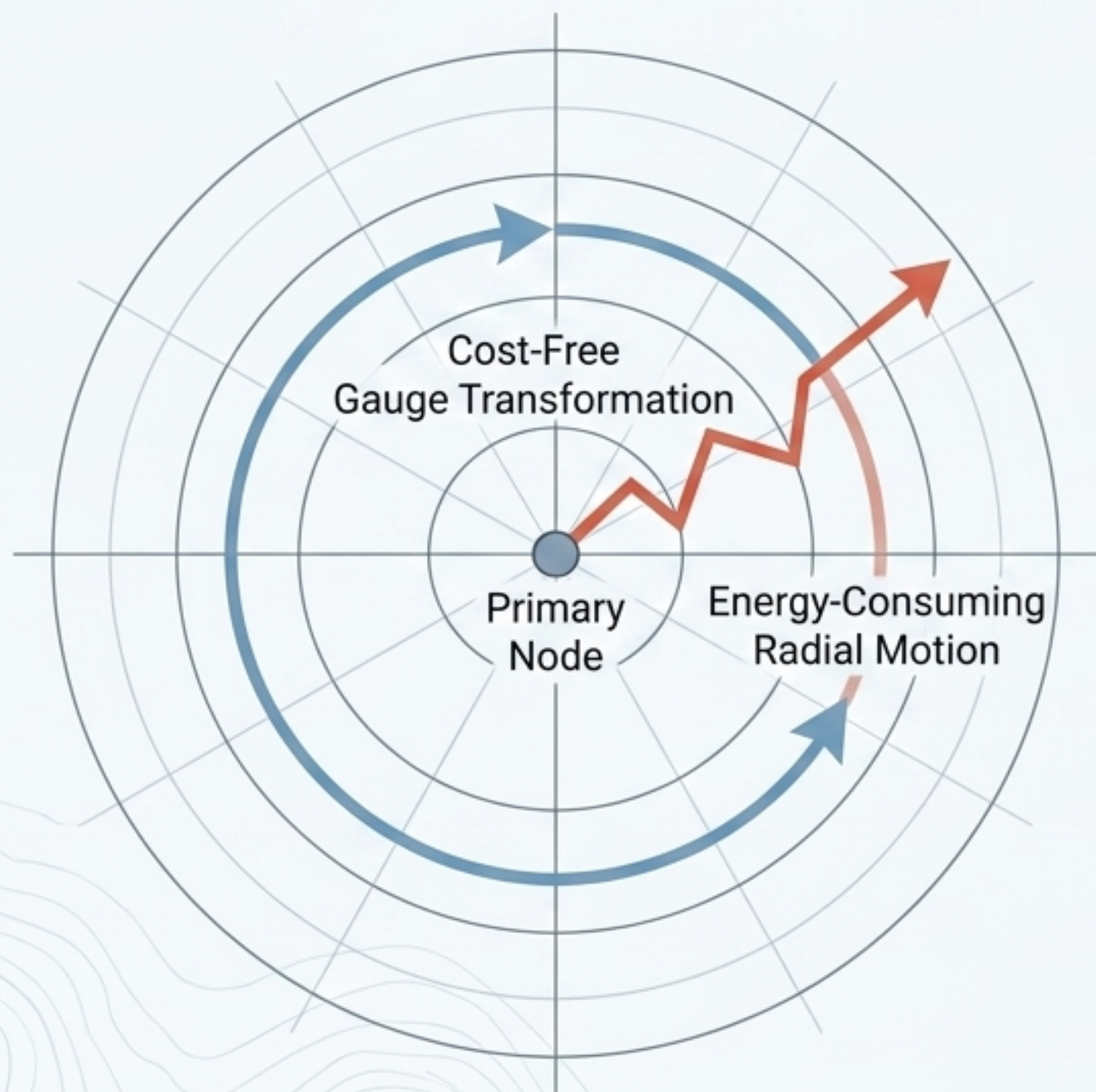
The Non-Holonomic Failure Mode

Two texts—"A caused B" and "B caused A"—occupy nearly identical coordinates in embedding space. Without geometric constraints, a trajectory can mathematically drift between them.

The Fix: Parallel Transport of Meaning

- The Semantic Connection ∇ defines "horizontal" motion (meaning preserved) vs. "vertical" motion (meaning altered).
- Admissible perturbations strictly require $\nabla_{\dot{D}}Q = 0$.
- If a closed loop returns with an altered meaning vector, the manifold has non-trivial Holonomy (Curvature Ω).

Noether invariants and semantic gauge freedom



Semantic Gauge Freedom

Substitutions like “researcher” $\leftrightarrow$ “scientist” are gauge transformations. They move smoothly along the orbit without consuming semantic budget because they do not alter the document’s projection onto critical axes.

Conservation Laws (The Invariant Sub-manifold F_Q)

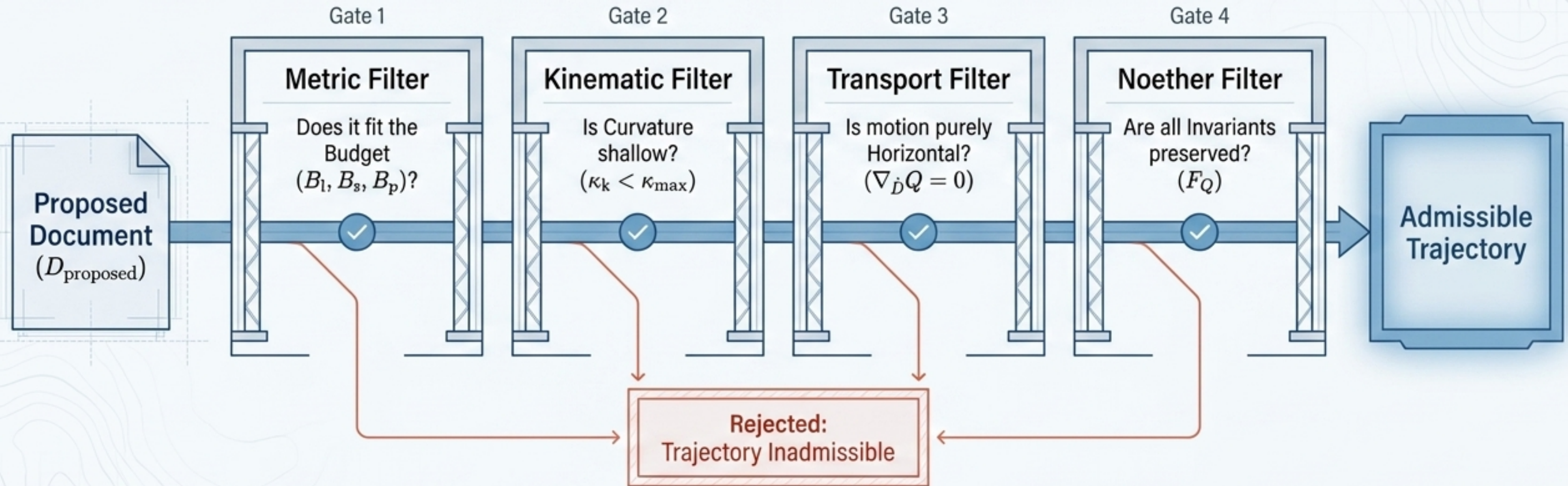
We enforce strict boundaries on conserved quantities $Q_i(D)$ to prevent structural collapse:

Thesis Projection: $Q = \langle \phi(D), \phi(\text{central claim}) \rangle$

Polarity: $Q = P(D) \in \{+1, -1\}$ (protected against negation flipping)

Causal Graph: $Q = G_D$ (protected up to graph isomorphism)

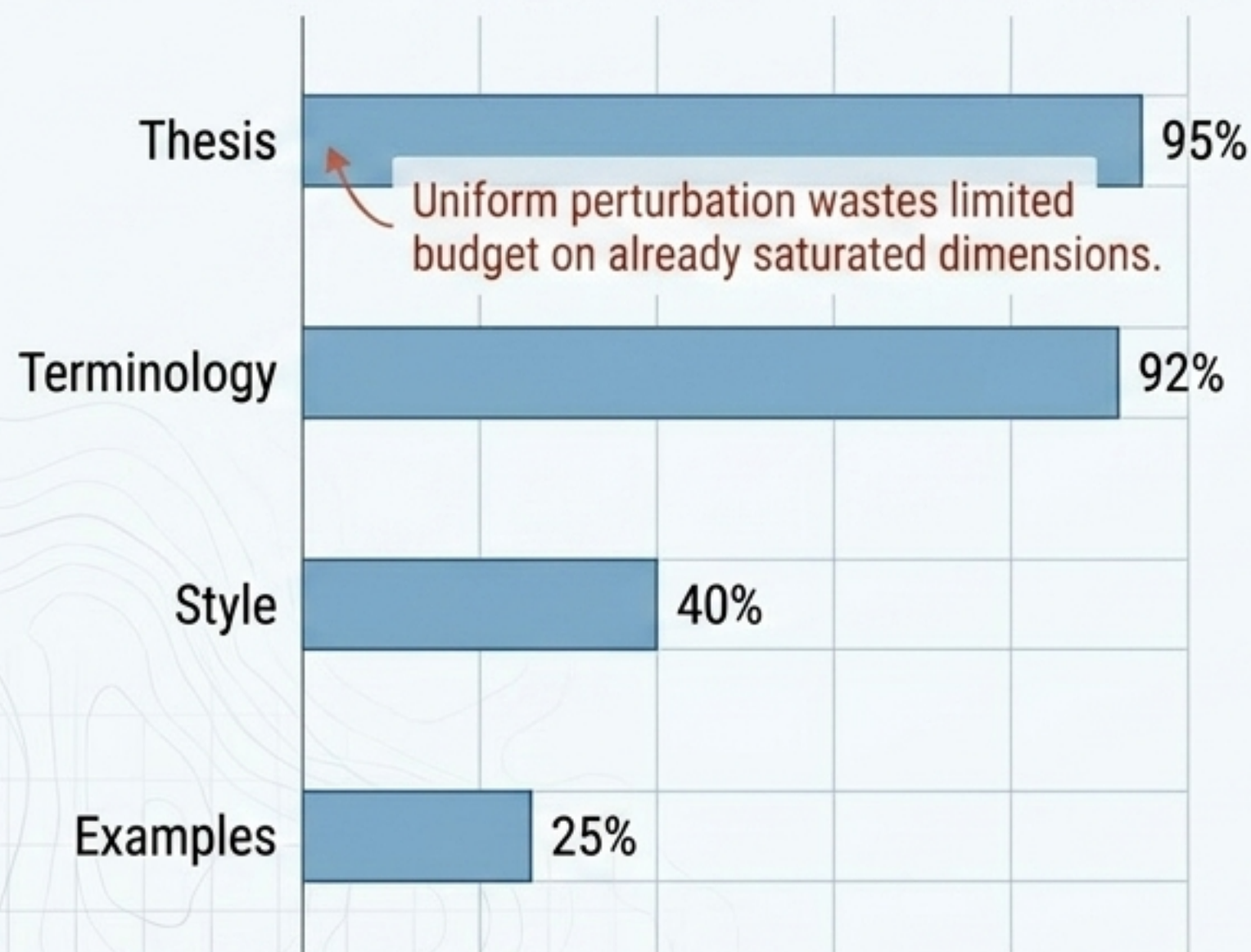
The anatomy of an admissible trajectory



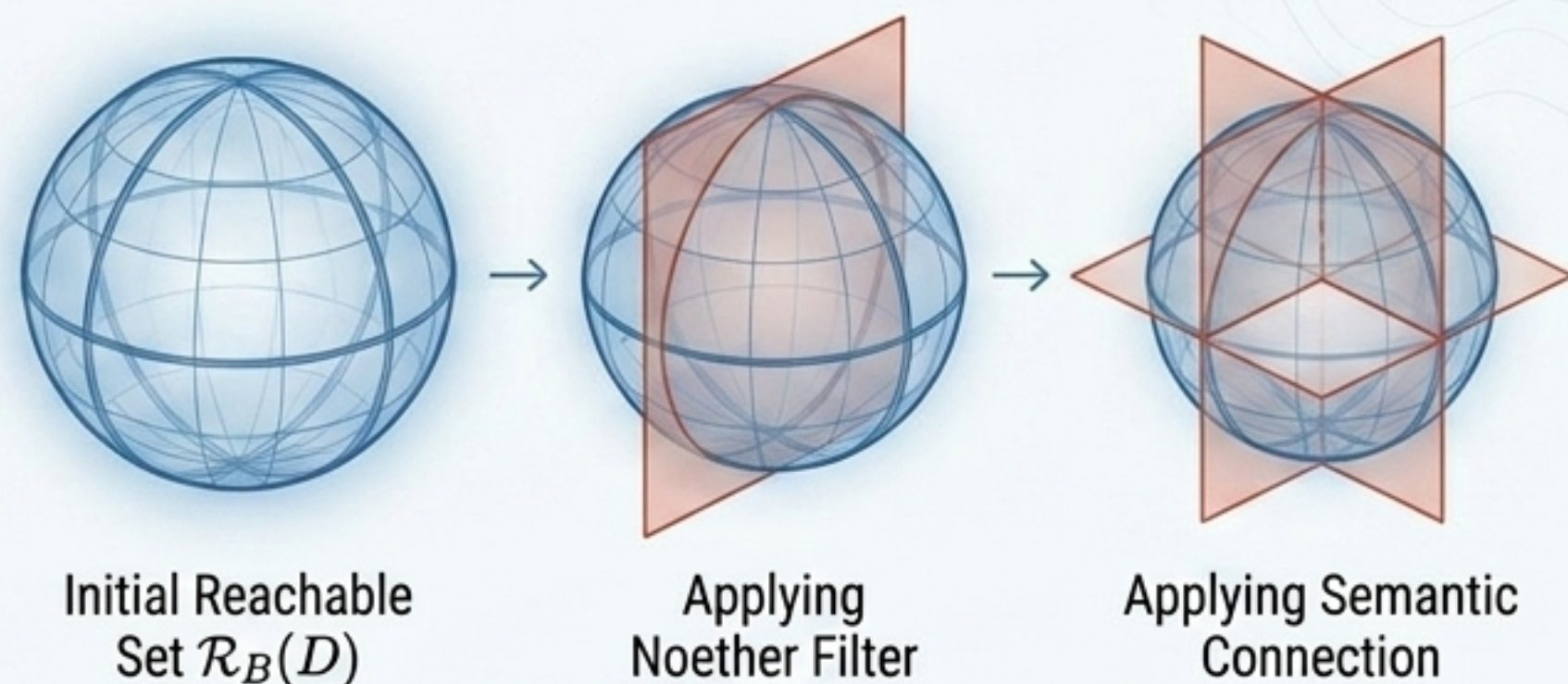
Synthesis Insight: Document perturbation is not a search for synonyms; it is the process of routing a trajectory through a sequence of strict geometric and kinematic filters to find the surviving sub-manifold.

Semantic Accessibility Entropy and Conceptual Bias

The Energy Problem (Semantic Spectral Bias)



Entropy Reduction via Conservation Laws

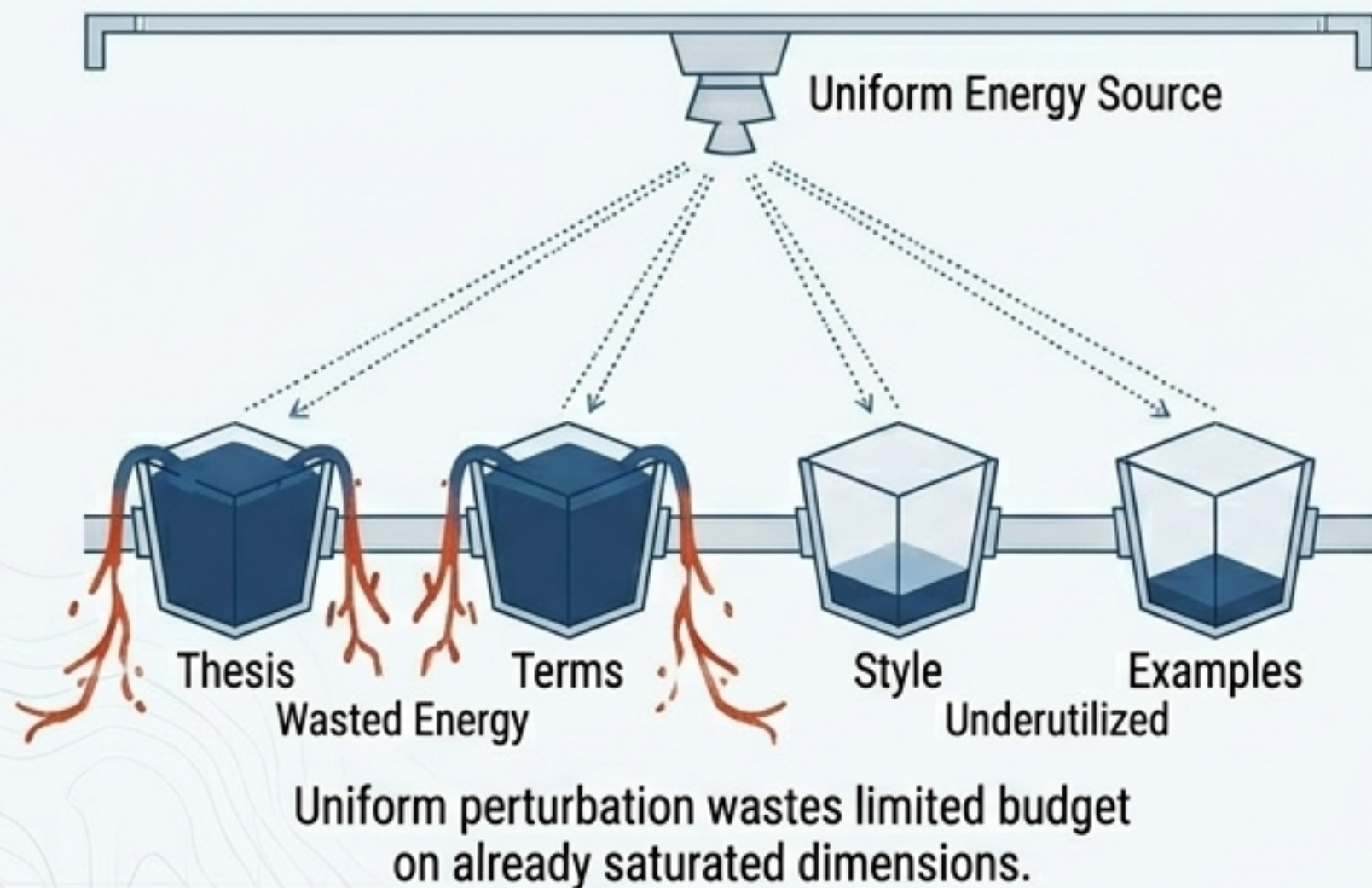


$$\text{Accessibility Entropy } S(D, B) = \log \text{Vol}_P(\mathcal{R}_B(D))$$

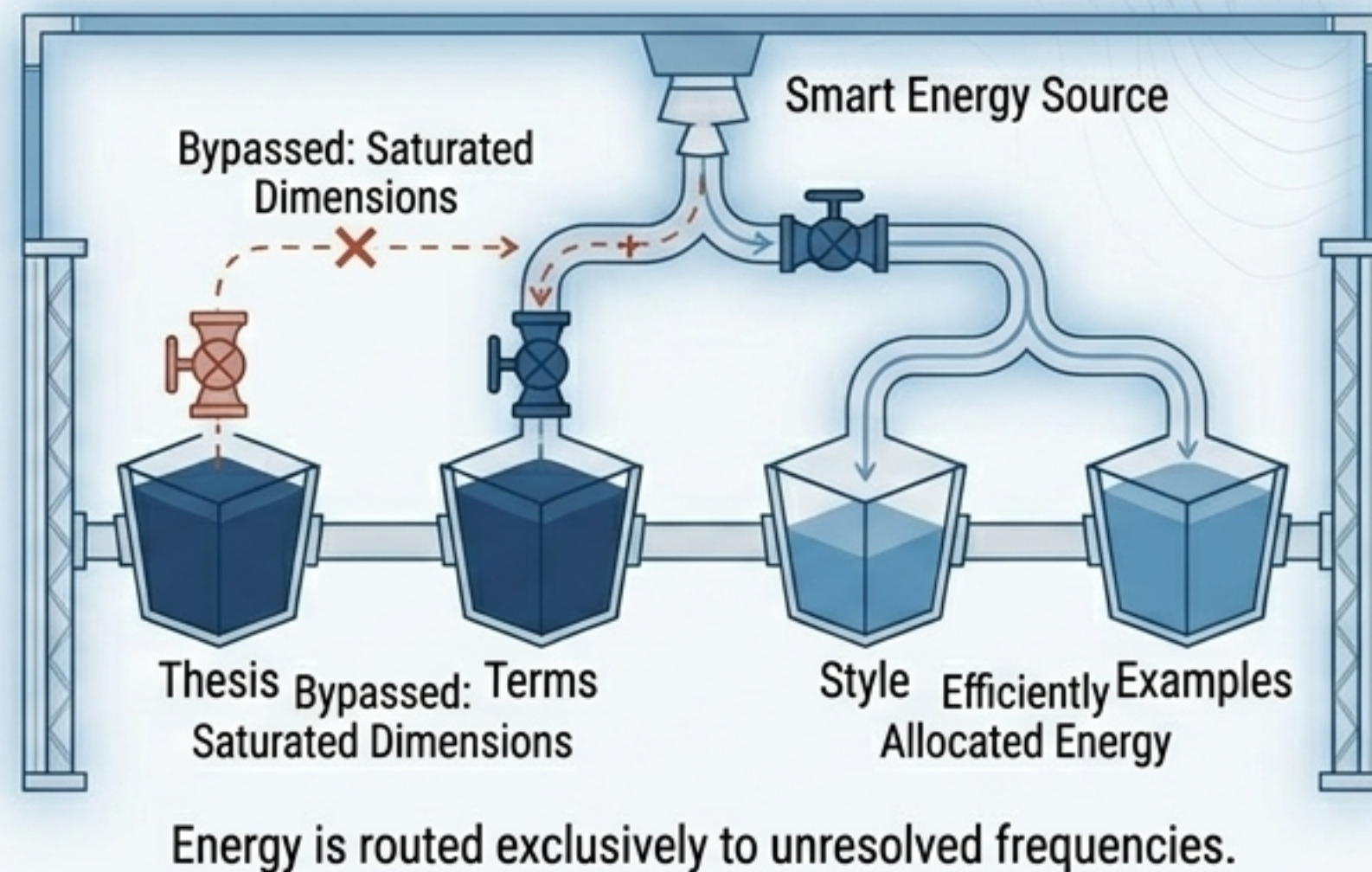
Proposition 7.10: Every conservation law applied geometrically shrinks the accessible volume, strictly reducing entropy.

Maximizing utility via Semantic Colored Noise

Traditional Paradigm (Semantic White Noise)



Theorem 7.5 Paradigm (Semantic Colored Noise)

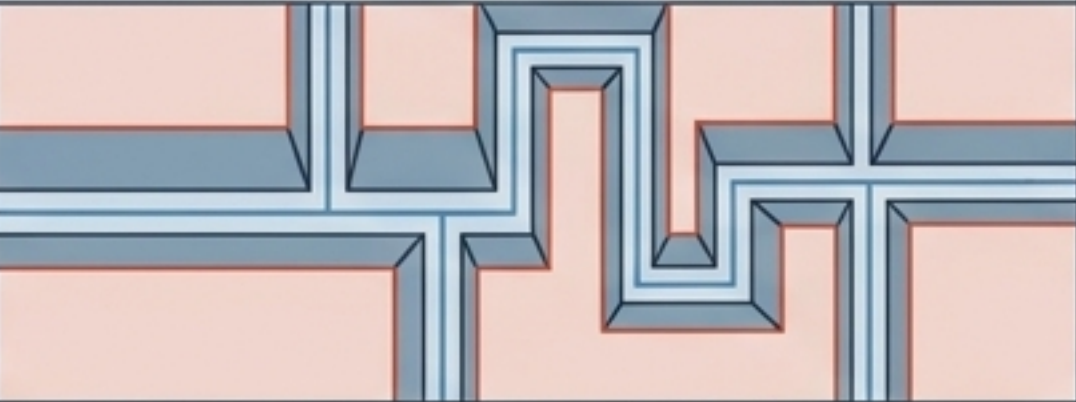


Math-to-Meaning Translation

Semantic utility experiences diminishing returns: $U_c(E) = \sqrt{E}(1 - \gamma(c))$ where $\gamma(c)$ is the completion field.

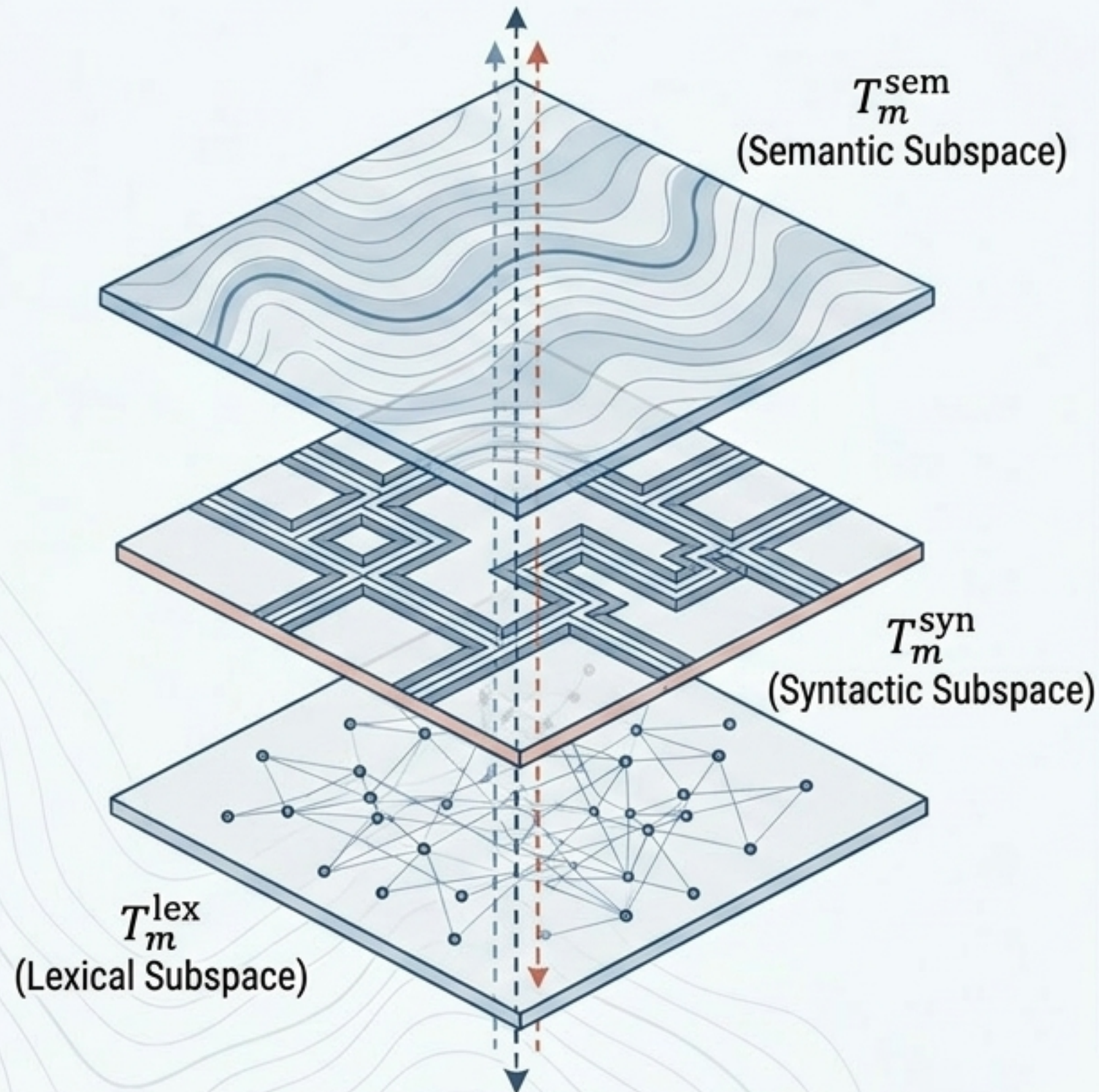
Theorem 7.5 (Budget Allocation Principle): To maximize semantic diversity under a strict distance budget, energy must be routed away from completed dimensions and directed exclusively into unresolved semantic frequencies.

Domain rigidity defines the local geometry of meaning

Academic	Legal	Literary
		
Metric Weight: Register shifting ($w_{\text{form}} \gg 1$)	Metric Weight: Deontic modality (<i>shall</i> vs. <i>must</i>)	Metric Weight: Tonal variance
Permitted Curvature (κ_{max}): Moderate	Permitted Curvature (κ_{max}): Narrow (strictly horizontal)	Permitted Curvature (κ_{max}): Wide
Connection Curvature ($\ \Omega_P\ $): Low	Connection Curvature ($\ \Omega_{\text{legal}}\ $): High	Connection Curvature ($\ \Omega_P\ $): Low
Resulting Rigidity $R(P)$: Moderate	Resulting Rigidity $R(P)$: Extreme	Resulting Rigidity $R(P)$: Minimal
Failure Mode: Losing formal tone.	Failure Mode: Loop accumulates deontic inversion.	Failure Mode: N/A (tolerates large semantic loops).

$$\text{Rigidity } R(P) = \frac{\|\Omega_P\|}{\kappa_{\text{max},P}}$$

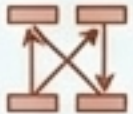
Multi-scale perturbation operators navigate specific tangent subspaces



Lexical Operators (Cost $\Delta \approx 0.001-0.012$)

 Synonym/register shifts via frequency-weighted WordNet. Selection interpolates frequency via $\alpha = 1 - 0.85t$.

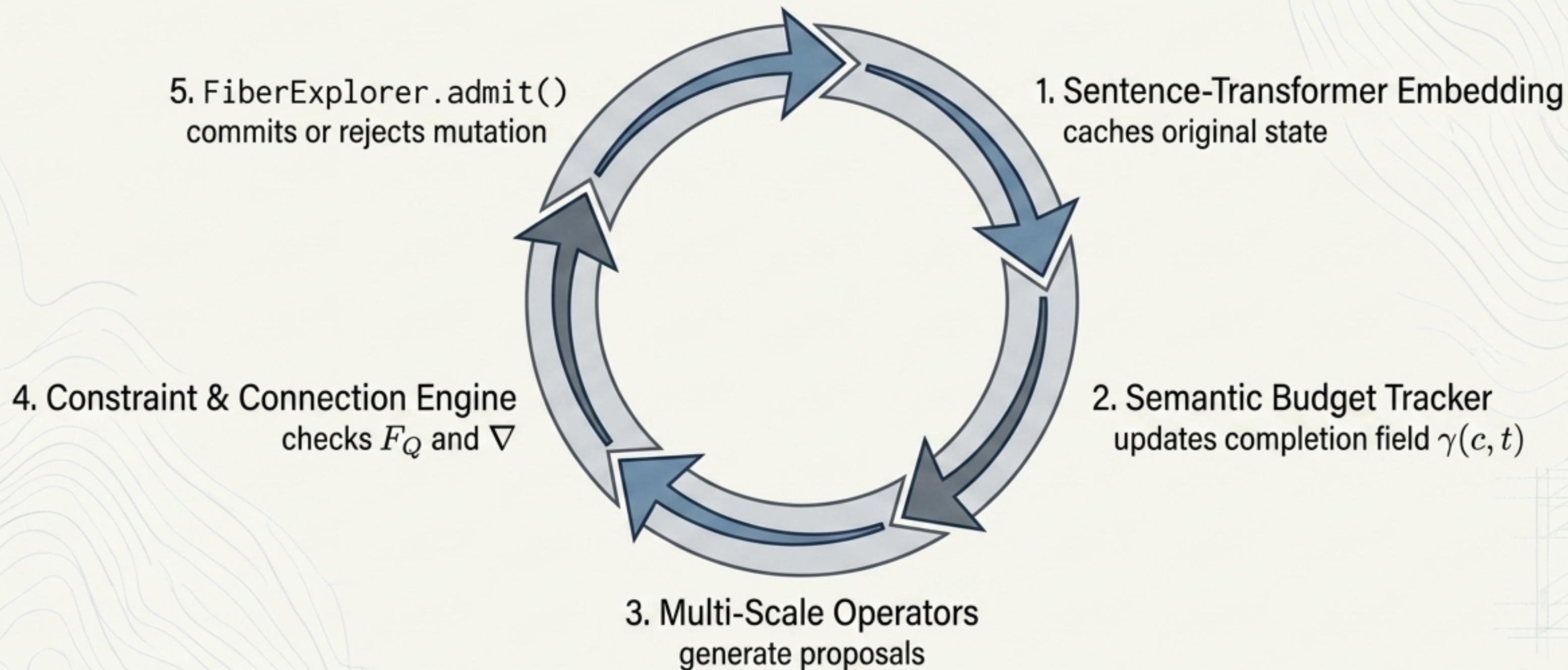
Syntactic Operators (Cost $\Delta \approx 0.008-0.020$)

 Clause reordering, active-passive. Highly susceptible to connection violations (e.g., inadvertently inverting causal graphs).

Semantic Operators (Cost $\Delta \approx 0.015-0.040$)

 LLM paragraph rewrites. Invoked only when remaining semantic budget ($B_p - \text{spent}_p$) securely exceeds safety thresholds.

The DocPerturb Control Loop

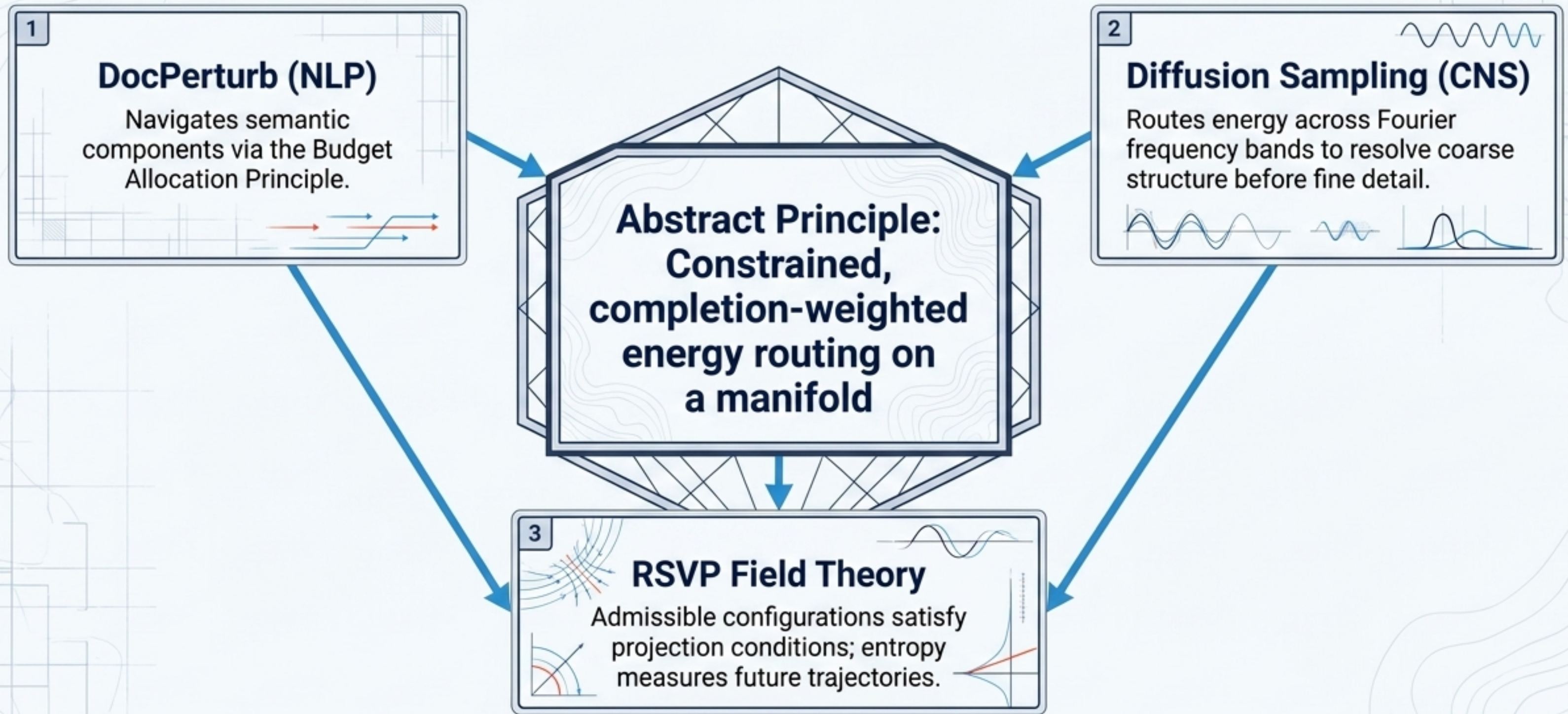


Framing: A 3,000-line implementation reframing text generation entirely as optimal control theory.

Efficiency: Sentence-level caching prevents recalculating the entire 200-mutation trajectory for every forward pass.

Optimization: Greedily optimizes colored noise budget allocation, verifying purely horizontal motion before committing.

The grand unification of document physics



Key Insight: The mathematics governing our semantic fibers are identical to leading frameworks across advanced physics and generative AI. They are unified by **finite-budget perturbation routing energy** toward unresolved dimensions.

Calibrated semantic distance applied to enterprise systems

RAG Robustness



Perturbing queries at mathematically calibrated distances ($B=0.05$ vs $B=0.15$) to test retrieval sensitivity without arbitrary semantic shifts.

Theory Version Control



Evolving legal statutes or software specifications where Noether invariants guarantee core claims are never inadvertently abandoned.

Plagiarism & Deduplication



Generating adversarial test cases with a formal mathematical guarantee that the underlying semantic identity has not been corrupted.

The Final Takeaway

Language is not a sequence of strings. It is a geometry. To control it, we must first map it.